

Case report: wall box with DLB-A1 controller

Charging current exceeds the wall box current limit, overload protection trips, no automatic resume after a DLB pause

1 Summary

An AmperePoint 11 kW three-phase wall box (Tuya device ID bfe26d78713489f8a6ulsj, firmware (V8.0.7)F1.3.6) was installed at a customer's house and worked correctly. After a DLB-A1 dynamic load balancing controller (board marking DLB CONTROLLER V1.4, 2026-1-21) was added to the installation, two problems appeared. On 26 September 2026 our installers reproduced both problems on site and recorded them on video. The same events are visible in the device event log on the Tuya platform. Everything described in this report is verified.

Problem 1 – priority (safety): in DLB mode the wall box does not respect its own current limit

With the wall box current limit set to **16 A** and the DLB Configured Supply Limit set to **18 A**, the vehicle drew **16.9 A**. When the wall box limit was lowered to **9 A** during charging, the wall box kept charging at **16.6 A / 7.1 kW** for several seconds and then tripped “*Current overload protection – automatic switch to low-current charging*”; charging stopped completely. With the customer's normal DLB setting of 20 A, the vehicle drew up to **19.1 A per phase (12.7 kW)**, although the wall box limit and the wall box rating are 16 A. **The on-board charger of the three-phase vehicle failed as a result: the vehicle became unresponsive and a mechanic was needed to restore it.**

Problem 2: after a DLB pause the wall box never resumes charging

When household load makes the DLB reduce the allocation below the 6 A minimum, the wall box pauses charging – correctly. When the load is removed, charging does **not** resume, not even after 5 minutes. It only restarts after the vehicle is unplugged and plugged in again.

Both problems occurred with **two independent DLB-A1 kits** and with **two different vehicles** (one with a two-phase and one with a three-phase on-board charger). Without the DLB, the same wall box on the same installation and the same socket works correctly. We therefore consider this a firmware issue of the wall box / DLB combination that most likely affects the whole batch, and we ask for an analysis and a corrected firmware. Evidence, log extracts and specific questions follow.

2 Equipment and configuration

Item	Details
Wall box	AmperePoint wall box, 11 kW, three-phase, maximum 16 A. Tuya device ID bfe26d78713489f8a6ulsj, product ID gbmngploofmhbjc, firmware (V8.0.7)F1.3.6, device info “Type B, AC 30mA + DC 6mA”, product variant 3, display language Polish. Supplied through a plug from the customer's socket outlet. The 433 MHz radio module is fitted on the internal connector, selector in position RF433 (Fig. 1).

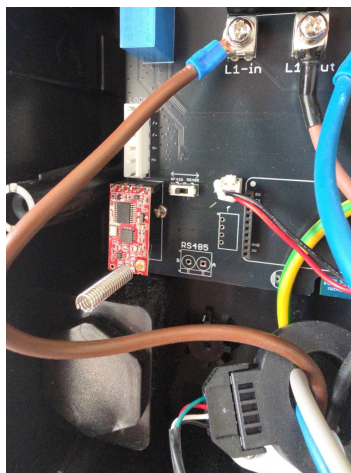
Item	Details
DLB controller	DLB-A1, board marking “DLB CONTROLLER V1.4 2026-1-21”. Three current transformers on L1/L2/L3 of the incoming supply, 433 MHz antenna with magnetic base on top of the distribution board (Fig. 1). Two independent kits (controller with accessories) were tested — the behaviour was identical with both.
Installation	Three-phase supply, main breaker 20 A. Customer's normal settings: DLB Configured Supply Limit (CSL) 20 A, wall box limit 16 A. Settings used during the tests are given with each problem.
Vehicles	Vehicle A: two-phase on-board charger (charges on L1 and L2 only). Vehicle B: three-phase on-board charger, 11 kW (16 A per phase). No charging-current limit is set in either vehicle. Both vehicles reproduce the problems.

What has been excluded

- **Installation:** before the DLB was installed, the wall box charged correctly on the same installation and the same socket.
- **Vehicle:** reproduced with two different vehicles (two-phase and three-phase on-board chargers).
- **Defective DLB unit:** reproduced with two independent DLB-A1 kits.
- **Damaged current transformer:** one CT had repaired (re-soldered) leads. After it was replaced, the current allocation became stable, but Problems 1 and 2 remained.



DLB-A1 controller in the distribution board, board marking DLB CONTROLLER V1.4, 2026-1-21



433 MHz radio module inside the wall box, selector in position RF433



433 MHz antenna with magnetic base on top of the distribution board; wall box breakers below

Figure 1: Hardware at the customer's site.

3 Problem 1 — current above the wall box limit, overload protection trips, charging stops

3.1 What the customer reported before our visit

- Since the DLB was installed, the wall box display shows the message “Ochrona przed przeciążeniem prądu – Automatyczne przełączenie na ładowanie niskim prądem” (Polish: “Current overload protection – Automatic switch to low-current charging”).
- When the wall box current limit was lowered while a vehicle was charging, the message appeared — the lower the limit, the sooner.
- With the limit set below 16 A, the three-phase 11 kW vehicle showed a charging power of **13 kW**, more than its on-board charger rating.

- The on-board charger of that vehicle failed. The vehicle was unresponsive and had to be restored by a mechanic.

3.2 Reproduction on site (video, 26 September 2026, 12:27–12:32 local time)

Vehicle A (two-phase). Settings: DLB Configured Supply Limit **18 A** (DLB display 018, Fig. 2), wall box limit **16 A**.

1. **1** Charging in progress. Wall box display: **DLB 16.9 A**, 7.3 kW, 225 V, limit 16 A. The current is already above the 16 A limit of the wall box.
2. **2** The wall box limit is lowered in the wall box menu from 16 A to **9 A** while charging. Display: DLB 16.9 A, 7.3 kW, limit 9 A.
3. **3** For several seconds the wall box keeps charging at **16.6 A / 7.1 kW** with a 9 A limit. The current is not reduced at all.
4. **4** After a few seconds the overload protection message appears. Display: DLB 0.0 A, 226 V, limit now **8 A** (lowered by the wall box itself). Charging has stopped completely — the wall box did not “switch to low-current charging”; it stopped.

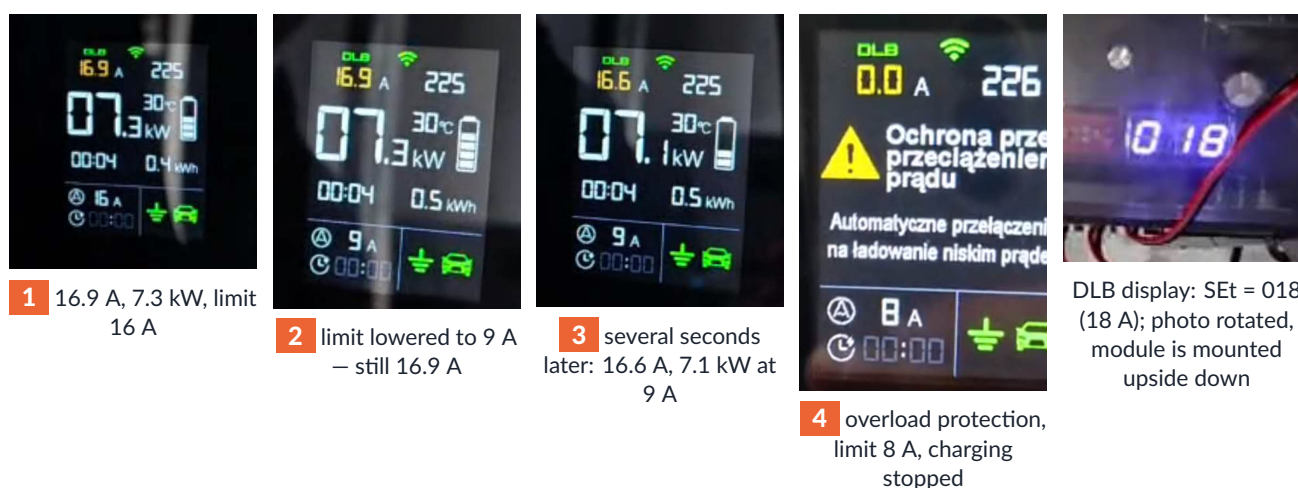


Figure 2: Problem 1 — frames from the installers' video.

3.3 The same minutes in the device event log (Tuya platform)

Times are local (UTC+2). Data point names are quoted as they appear in the Tuya log.

Time	Event
12:27:14	Wall box online after power-on; 设置充电电流 (current limit) = 16; 12:27:20 工作状态 (work state) = 300 (charging)
12:29:11 – 12:31:10	指标信息 (measurements): L1 17.0 A, L2 15.3 A, L3 0 A, 7.3 kW — steady
12:31:57 – 12:31:59	设置充电电流 reported by the device: 6 → 7 → 8 → 9 (limit changed in the wall box menu while charging; state still 300)
12:32:09	报警信息 (alarm) {"t": "2026-09-26 12:32:05", "v": 400}; 工作状态 = 400 ; 调试用工作状态 = 发生故障 (fault); 设置充电电流 = 8 ; measurements 0 A
12:32:30	充电记录 (session record): 12:27–12:32, 287 s

The log matches the video exactly: 7.3 kW, limit 9 A, then the alarm with the limit lowered to 8 A. The wall box's current measurement (17.0 A) also agrees with the DLB's own measurement shown on the display (16.9 A).

3.4 The same behaviour in other sessions on the same day (log)

The current exceeded the wall box limit **that had been set before the session started** – this is not only an effect of changing the limit during charging:

Time	Wall box limit	Current L1 / L2 / L3	Power	Context
11:39:33 – 11:43:30 and 11:44:13 – 11:48:25	16 A	18.4–18.7 / 18.8–19.1 / 18.2–18.5 A	12.4– 12.7 kW	Vehicle B (three-phase), customer's DLB setting CSL 20 A; about eight minutes in total (interrupted by a power cycle at 11:43), no alarm
11:51:40	15 A	20.2 / 19.1 / 0 A	8.8 kW	Vehicle A; alarm 2 s later
11:53:41	16 A	20.8 / 19.1 / 0 A	8.9 kW	Vehicle A, 36 s after power-on; alarm at the same second
11:59:29	13 A	16.4 / 16.4 / 15.8 A	10.9 kW	Vehicle B, 50 s after power-on; alarm 2 s later
12:24:30	10 A	16.7 / 15.6 / 0 A	7.2 kW	Vehicle A, 14 s after power-on; alarm 3 s later
12:57:08	16 A	18.7 / 18.8 / 18.2 A	12.6 kW	Vehicle B, 2 min after power-on

The 18–19 A per phase measured by the wall box is the same value that the customer saw on the vehicle as “13 kW” – two independent measurements agree.

3.5 Overload protection – all 12 events in the log

Data point 报警信息 with value 400 together with 工作状态 = 400 and 调试用工作状态 = 发生故障. “Current at alarm” is the last measurement before the alarm; after the host-heartbeat data point (上位机心跳) switches to off, the wall box reports measurements only every 60 s, so for some alarms the last measurement is older.

#	Time	Limit before → after	Current at alarm (L1/L2/L3), % of limit	Context
1	11:50:40	16 → 15	9.1 / 6.4 / 0 A five seconds earlier, still rising	new session, 10 s after start
2	11:51:43	15 → 15	20.2 / 19.1 / 0 A – 135 %	vehicle resumed by itself 41 s after alarm 1
3	11:53:41	16 → 15	20.8 / 19.1 / 0 A – 130 %	36 s after power-on
4	11:56:18	13 → 13	9.3 / 9.9 / 0 A five seconds earlier, still rising	limit 13 set in the menu before the session; alarm 10 s after start
5	11:59:31	13 → 12	16.4 / 16.4 / 15.8 A – 126 %	52 s after power-on, Vehicle B
6	12:05:56	7 → 6	18.2 / 18.3 / 17.7 A – 261 %	limit lowered 16 → 7 in the menu during charging, 10 s earlier
7	12:22:29	10 → 9	12.9 / 11.8 / 0 A – 129 %	limit lowered 16 → 10 in the menu during charging, 10 s earlier
8	12:24:33	10 → 9	16.7 / 15.6 / 0 A – 167 %	16 s after power-on

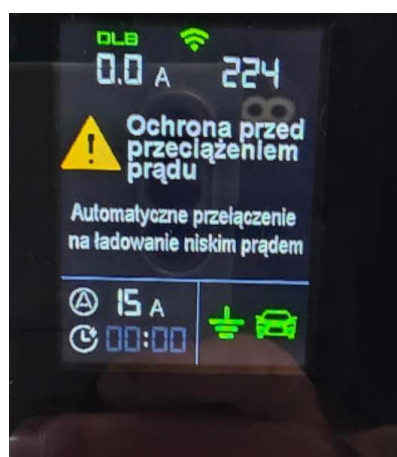
#	Time	Limit before → after	Current at alarm (L1/L2/L3), % of limit	Context
9	12:26:33	16 → 15	8.8 / 9.1 / 0 A five seconds earlier, still rising	27 s after power-on
10	12:32:09	9 → 8	17.0 / 15.3 / 0 A one minute earlier	limit lowered 16 → 9 in the menu during charging (the video)
11	12:34:27	8 → 7	1.1 / 1.3 / 0 A 24 s earlier	new session with limit 8
12	13:00:13	16 → 15	10.2 / 9.6 / 0 A 35 s earlier	91 s after power-on

What the log shows about the protection:

- All 12 alarms occurred **while charging** (state 300). None occurred while idle or with a vehicle connected but not charging.
- Reaction: charging stops within about 1 s (measurements 0 A), state 400, and the wall box **lowers its own limit by 1 A** (16→15, 13→12, 10→9, 9→8, 8→7, 7→6). This lowered value is not stored: after a power cycle the previous limit returns.
- In seven cases the wall box returned to “vehicle connected” by itself after 10–55 s; in the other five the power was switched off 24–141 s after the alarm while the wall box was still in the fault state. When the vehicle restarted charging by itself after alarm 1, alarm 2 followed 21 s later (20.2 A with limit 15 A).
- Where the current at the moment of the alarm is known, it was always $\geq 126\%$ of the limit; 119 % (19.1 A at limit 16 A) was tolerated for about eight minutes without an alarm. The lower the limit, the sooner the alarm — exactly what the customer reported.
- Lowering the limit in the wall box menu during charging (12:05:45, 12:22:17, 12:31:57) produced an alarm about **10 s after the last key press**, and the current was **not reduced** before the alarm.

Why this is a safety issue

The vehicle drew 19.1 A per phase on an installation with a 20 A main breaker and with a wall box rated 16 A. Any additional household load would trip the main breaker. The on-board charger of the three-phase vehicle has already failed.



Customer's photo of the message: *Ochrona przed przeciążeniem prądu – Automatyczne przełączenie na ładowanie niskim prądem*; limit shown 15 A (the wall box lowers the limit by 1 A at each alarm, see 3.5)



Normal charging with the DLB: DLB 12.1 A, 9.0 kW, limit 16 A — the DLB field shows the DLB-measured current



Idle, DLB paired (green icon): DLB 0.0 A, 243 V, limit 16 A

Figure 3: Customer's photos of the wall box display.

4 Problem 2 — no automatic resume after a DLB pause

4.1 Reproduction on site (video, 26 September 2026, 12:36–12:44 local time)

Vehicle B (three-phase). Settings: the DLB Configured Supply Limit was **deliberately set to a low value** so that a household load would push the allocation below the 6 A minimum; wall box limit 16 A.

1. **1** Charging at **DLB 10.8 A**, 7.4 kW, 226 V, limit 16 A (log 12:36:13–12:37:34: 10.5–10.8 / 10.7–11.0 / 11.1–11.6 A, 7.4 kW).
2. **2** A kettle is switched on as additional household load. The DLB reduces the allocation below 6 A and the wall box **pauses charging** — **this is correct**. Display: DLB 0.0 A, 0.0 kW, 228 V, limit 16 A, vehicle still connected (log 12:37:53: 工作状态 = 200, vehicle connected, not charging).
3. **3** The kettle is switched off. **The wall box does not resume charging**. The installers waited 5 minutes — still paused. In the log the wall box stayed in state 200 from 12:37:53 until it went offline at 12:44:21; no charging was recorded.
4. **4** Only after the vehicle is **unplugged and plugged in again** does charging restart.



1 charging at DLB 10.8 A, 7.4 kW, limit 16 A



2 paused after the kettle was switched on: DLB 0.0 A, 0.0 kW, vehicle connected — and it stays like this

Figure 4: Problem 2 — frames from the installers' video.

The DLB-A1 manual (section 6.2, *Paused*) states: “Recovery is automatic when headroom returns.” This does not happen.

5 Observed behaviour versus documented behaviour

Function	DLB-A1 Quick User Manual (Rev. 20251109)	Observed
Upper bound of the charging current	“each wall box is capped by its own requested current”	Current exceeds the wall box limit: 16.9 A at 16 A; 16.6 A at 9 A; 20.2 A at 15 A; 16.7 A at 10 A; 19.1 A at 16 A
Usable ceiling	“90 % of the Configured Supply Limit (10 % buffer)”	With CSL 18 A the vehicle drew 16.9–17.0 A; with CSL 20 A up to 19.1 A — no 10 % buffer visible
Reduction when headroom drops, pause below 6 A	reduces, keeps ≥ 6 A where possible, otherwise pauses	Works (Problem 2, step 2)

Function	DLB-A1 Quick User Manual (Rev. 20251109)	Observed
Resume after pause	"Recovery is automatic when headroom returns"	No resume, not even after 5 min; vehicle must be re-plugged
Overload protection message	"Automatic switch to low-current charging"	Charging stops completely; limit lowered by 1 A; the next start trips again

6 Our interpretation – hypothesis, for your verification

The following is our reading of the evidence; it is **[hypothesis]** and we ask you to confirm or correct it.

1. In DLB mode the current allowance signalled to the vehicle on the Control Pilot appears to follow the DLB allocation only and is **not limited by the wall box's own current limit**. The observed allowance is close to the Configured Supply Limit itself (CSL 18 A → 16.9–17.0 A; CSL 20 A → 18.7–19.1 A; vehicles typically draw 95–99 % of the allowance), which also suggests that the 10 % safety buffer is not applied in this firmware.
2. The overload protection compares the **measured** current with the wall box limit and trips at roughly 120–125 % of the limit. It then lowers the limit by 1 A and stops charging. Because the limit does not bound the current in DLB mode, the lowered limit does not reduce the current, so the alarm repeats at the next start – and the lower the limit, the sooner it trips.
3. When the limit is lowered during charging, the new value is applied to the protection threshold immediately while the CP allowance stays at the DLB value. The vehicle is never asked to reduce its current, and the protection trips about 10 s later.
4. Problem 2 is separate: after a pause caused by an allocation below 6 A, the wall box (or the DLB) does not restart the session when headroom returns.

Expected behaviour: CP allowance = min(wall box current limit, DLB allocation); a limit change during charging first lowers the PWM duty cycle and gives the vehicle the standard time to comply before any protection is evaluated; automatic resume after a pause.

7 Questions and requests

1. In DLB mode, is the CP allowance calculated as **min(wall box current limit, DLB allocation)**? Our observations indicate that it is not.
2. Which allocation formula does DLB-A1 firmware V1.4 actually implement? The manual states $90\% \times \text{CSL}$ minus the highest CT-measured phase current; we observed an allowance close to the CSL without the 10 % buffer.
3. What is the trip condition of the overload protection (alarm 400), and what exactly does "automatic switch to low-current charging" do? We observed: stop, limit –1 A, no resume by the wall box.
4. When the current limit is changed during charging, does the wall box lower the CP allowance first and wait for the vehicle to comply before evaluating the protection?
5. Why does the wall box not resume charging after a DLB pause when headroom returns? Is a re-plug of the vehicle required by design?
6. Please provide a corrected firmware for the wall box (currently (V8.0.7) F1.3.6) and/or for the DLB-A1 (V1.4), and tell us which firmware versions and production batches are affected, so that we can inform other customers.
7. For diagnostics: could the DLB allocation (the target current sent to the wall box) be reported as a data point in the Tuya log? At present the log contains no DLB-related data point.

Interim measure we give to customers

Until a corrected firmware is available: operate the wall box without the DLB (it works correctly without it), or set a charging-current limit in the vehicle equal to the wall box limit. The vehicle uses the lower of its own limit and the CP allowance.

8 Attachments and available material

- Installers' video of 26 September 2026 (Problem 1 and Problem 2) – available on request; frames in Fig. 2 and Fig. 4.
- Customer's photos (Fig. 1, Fig. 3).
- Full Tuya device event log of bfe26d78713489f8a6u1sj for 26 September 2026, 10:43–14:26 local time (1 761 events) – export available on request. All times in this report are local (UTC+2); the log itself is stored in UTC.
- Photos of the charging inlets of both vehicles (Fig. 5).



Vehicle A – two-phase
on-board charger



Vehicle B – three-phase
11 kW on-board charger

Figure 5: Vehicles used in the tests.

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